SHINVA 新华医疗

Automatic Washer-disinfector

Installation Manual

Rapid-M-320

山东新华医疗器械股份有限公司
SHINVA MEDICAL INSTRUMENT CO.,LTD.

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1. The conventional safety regulations

1.1 Regulation requirements

If the content contained in this manual conflicts national or local regulations, the health department regulations and other policy documents, conform them; otherwise conform this manual; this manual is used to guide the design and installation of CSSD and Rapid-M-320 washer disinfector, ensure that the equipment is correctly installed and fixed, and conveniently used and maintained.

1.2 The installation environment:

- Prohibit to install in environment with heavy dust, dirt or containing conductive particles, corrosive gas, combustible gas ;
- Prohibit to install in environment at high temperature and high humidity or easy to be wet;
- Prohibit to install in environment in a strong magnetic field ;
- The ground (equipment supporting feet, the relative position please see the installation diagram) maximum bearing pressure: 0.5Mpa;

1.3 The installation space

The minimum space for the normal operation of the equipment: w * D * h =1410 x 4000 x2020mm;

1.4 Precautions before installation

Before the equipment installation and during the installation process, prevent the collision and damp. It is not allowed to turnover the equipment in any situation. In the plane place it can be transferred by its bottom wheels or forklift. If crane must be used (such as installation is located on upper than ground floor), the crane sling can't generate pressure at anywhere of the equipment except its bottom;

After the equipment is put in site, adjust the feet bolts of equipment to horizontal level, and before the equipment is powered on, it must be ensured to expose for 24 hours to remove its moisture;

The technical parameters or same kind digital contained in this manual, if no special instructions, refers to the single equipment parameters.

2. The Installation Mode

2.1 Embedded-type installation

If using embedded installation, reserved 300mm maintenance should be kept in each side of

equipment. If many equipment are installed side by side, each set shall is adjusted to the same height, and between each equipment and 300mm maintenance space should be preserved away from the wall. As shown in figure 1:

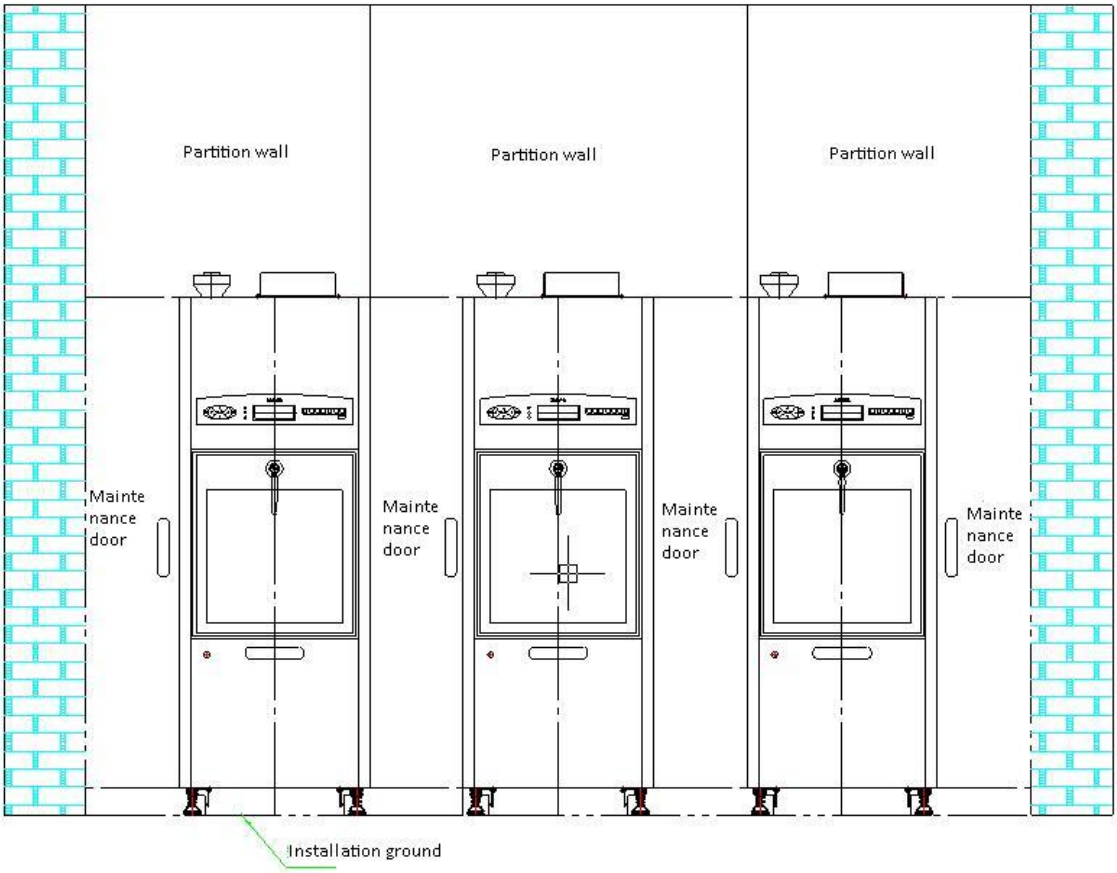


Fig 1

2.2 The external dimension of the equipment

Equipment installation external dimensions: width × depth x height =710 x 760 x 1930 (mm).

2.3 Precautions while Installation

- The partition wall or any other decoration material cannot be fixed on the equipment in any way, to ensure that the equipment can be easily removed or replaced in the face of major damage or failure.
- While installing the equipment , must adjust the transporting trolley and the orbit in the equipment to a uniform height, which prevent that the trolley is unable to adjust after the ground is made well;
- The equipment should be firmly fixed, to prevent the equipment from moving because of vibration in its running process;

Notice: if the equipment is fixed with bolts, screws around its two sides, after the equipment is in place, the fixed bolt or screw must be removed!

3. Control power supply and power cut off device

3.1 Power requirements

380VAC, 50Hz ， 22.5KW

3.2 The power cut off device

In the power supply cables, each piece of equipment must be equipped with a separate to to cut off the power supply. The switch can be installed on the wall and operated conveniently.

3.3 Electrical connections

- It is required to install the power switch box near the wall; switch box must be matched with a three-phase knife switch (or circuit breaker).
- In order to ensure personal safety and equipment safety, an earth wire must be laid, among the control cables a cable marked “earth” must be connected to the ground reliably. All the pipes and cables shall be kept vertical and horizontal and fixed.
- If the equipment is installed near the other equipment, it should be able to find the each breaker switch for each machine conveniently, so in an emergency user can accurately close corresponding power switch.

4. Adjust the equipment height

After the equipment is put in place, its height and level of the equipment and should adjusted;

By adjusting the equipment foundation foot bolt to adjust the height of equipment, it is appropriate to keep a distance of 1930 ~ 2000mm from the top equipment to the ground and based on this height adjust the horizontal level;

If use buy a transport cart, should adjust the trolley height: open the front door and back door of the equipment, put the washing rack, and adjust the transport cart to the height that the cleaning track can be smoothly move in chamber direction

5. The connection of pipelines

5.1 The water inlet pipeline

5.1.1 Equipment water inlet

The equipment consists of 3 water inlet (softened water inlet A and B, pure water inlet C), connected separately to the cleaning tank and the preheating water tank; Figure 2 is a top view of equipment.

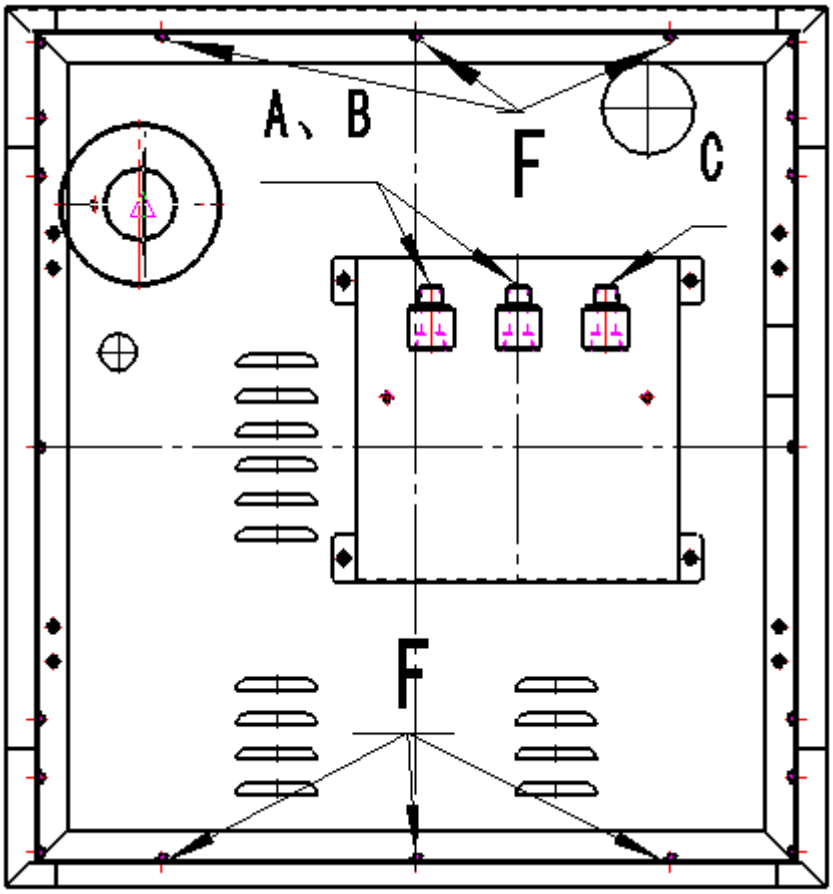


Fig 2

Notice: After the equipment is in put in place, remove the screws (total six pieces) in F point shown in Figure 2. 0therwise, after the equipment room decoration is completed, the front and back control cover and side cover cannot be open.

5.1.2 Water inlet size

The water inlet pipe joint is 1/2 “diameter inside screw stainless steel tube

5.1.3 Water quality requirement

Water quality requirement: it is recommended to connect all with pure water or distilled water (conductivity is ≤15μS/cm)! If user cannot meet the requirements, inlet A, B can be connected with softened water, and its hardness does not exceed 7°d .

Notice: the factors influencing the cleaning effect are from many aspects. If the water quality does not reach the requirement, after cleaning and drying the device, its surface easily to generates white spots!

5.1.4 Water source pressure requirement

Water pressure: 0.2~0.5Mpa;

The inlet pipe of each set of equipment should have separate control valve, which is

installed in a position convenient for operating;

Note: separated control valve can ensure to cut off the water pipe on the need to maintain one set of equipment, but does not affect other equipment running!

The water inlet pipeline joint position is as shown in Figure 2: (in diagram labeled A, B, C is water inlet).

5.2 Drainage and exhaust pipe connection

5.2.1 Drainage Pipe sizes

Drainage pipe joint D is ϕ50 stainless steel, connecting stainless steel pipe or temperature resistant hose to the total drainage pipeline. Inside diameter of the drainage tube is Φ46mm; the total drainage tube diameter of less than two sets of equipment (including 2 sets) shall be not less than 70mm, and of three or four sets of equipment not less than Φ100mm, and of three or four sets of equipment not less than Φ125mm

5.2.2 Exhaust pipe size

The exhaust pipe join E is "◎" type double-layer annular pipe, and its outer ring is Φ140mm stainless steel and inside Φ63.5mm stainless, connecting the external stainless steel tube or temperature resistant hose with inside diameter Φ80 - Φ130mm (if multiple equipments are installed, the total exhaust pipeline section area S_{total}≥ equipmentsx40cm^2) to the open air, the steam in exhaust gas will be condensed and goes along the exhaust pipe into the "◎" double-layer annular pipe's bell-shape joint, and then through the bell-shape joint discharged into the drainage pipeline.

Notice: because the water and gas discharged can reach highly to 90 ℃, so the drain pipe must be high temperature resistant, non-toxic!

The exhaust pipe is only required to covered Φ63.5mm joint of the equipment and cannot be tightly connected, the outdoor terminal of the external exhaust pipe should be matched with a ventilation fan (air volume of the single equipment is ＞10 m^3/ minute);

Notice: external exhaust pipe cannot be closely connected with equipment exhaust pipe, and the outer end must be installed with a fan; otherwise the temperature of partition wall is easily lead to an over high value, which affecting the normal running of equipment!

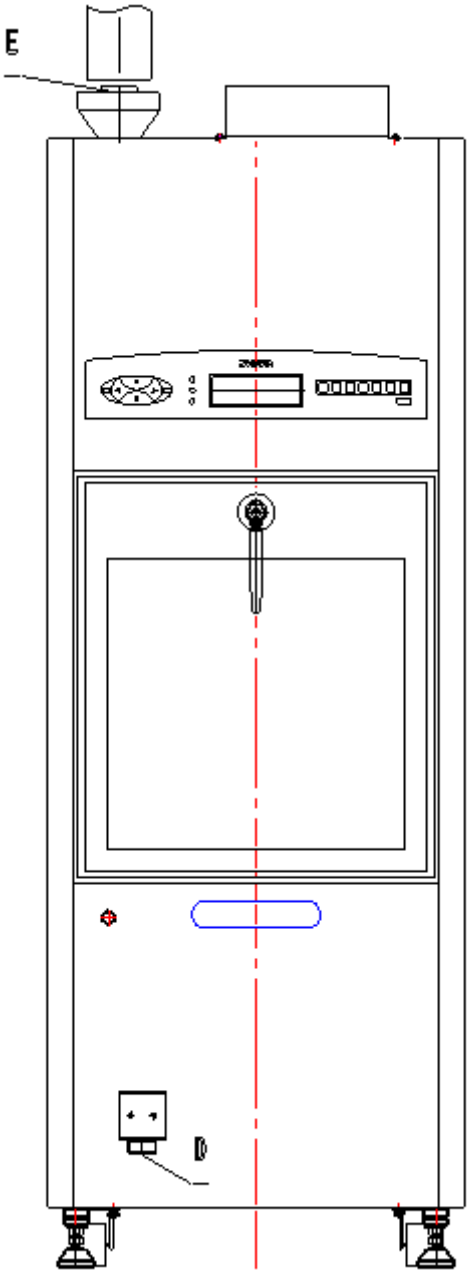


Fig 3

6. Equipment partition

6.1 Double side partition

Equipment adopts the front and rear double side partition;

6.2Partition form

The partition is only done after the equipment installation and pipeline connection is completed; partition wall can not be directly fixed on the equipment, if necessary, can be supported by steel structure to; Fig 4 is the positive view after partition:

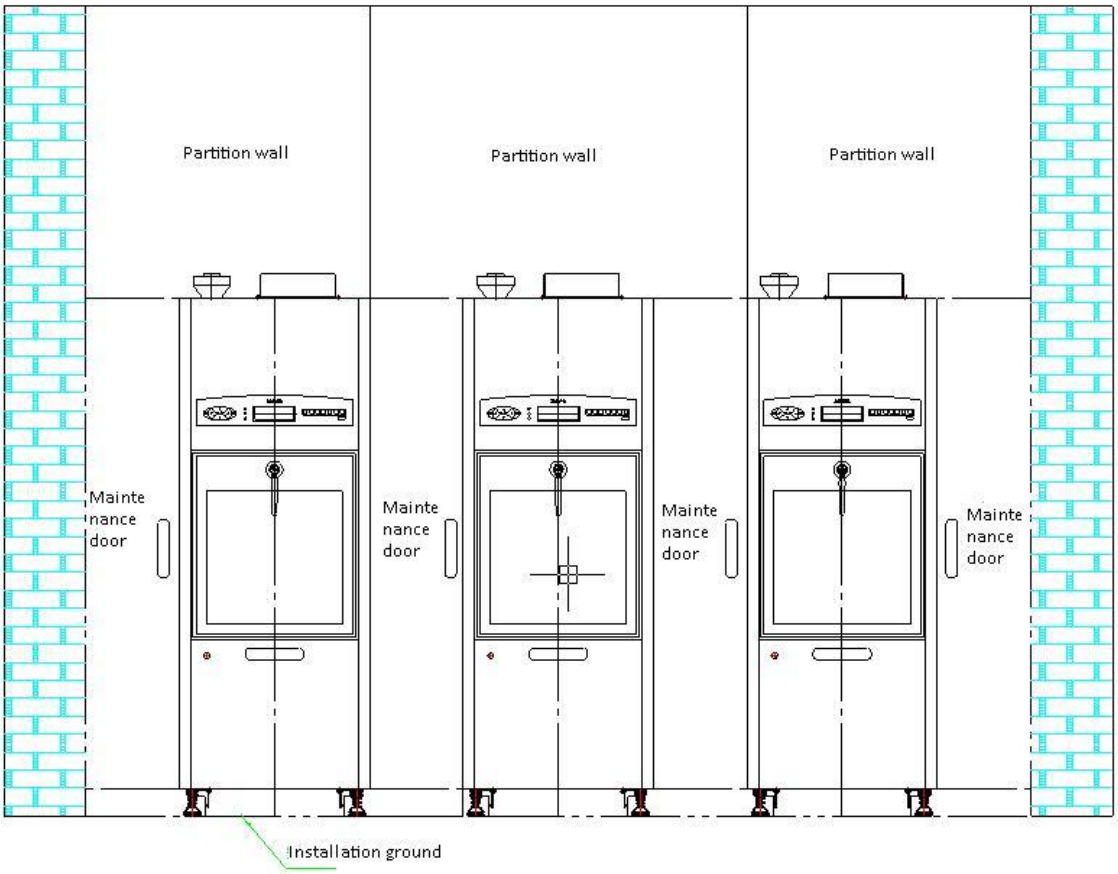


Fig 4

The front and back side of the partition should be in the same level with the front and back side of the equipment. The left side view is shown in Figure 5 (not including the pipeline part):

Notice: the partition walls or other decorative materials cannot fix on the equipment in any way, otherwise it will be difficult to repair or replace the equipment once it suffered major damage or failure!

Notice: between the inside and outside an exhaust fan must be installed to ensure the partition air is always in the state of negative pressure, otherwise the air humidity in the partition will too large that will easily cause electrical failure or reduce the use life of the equipment!

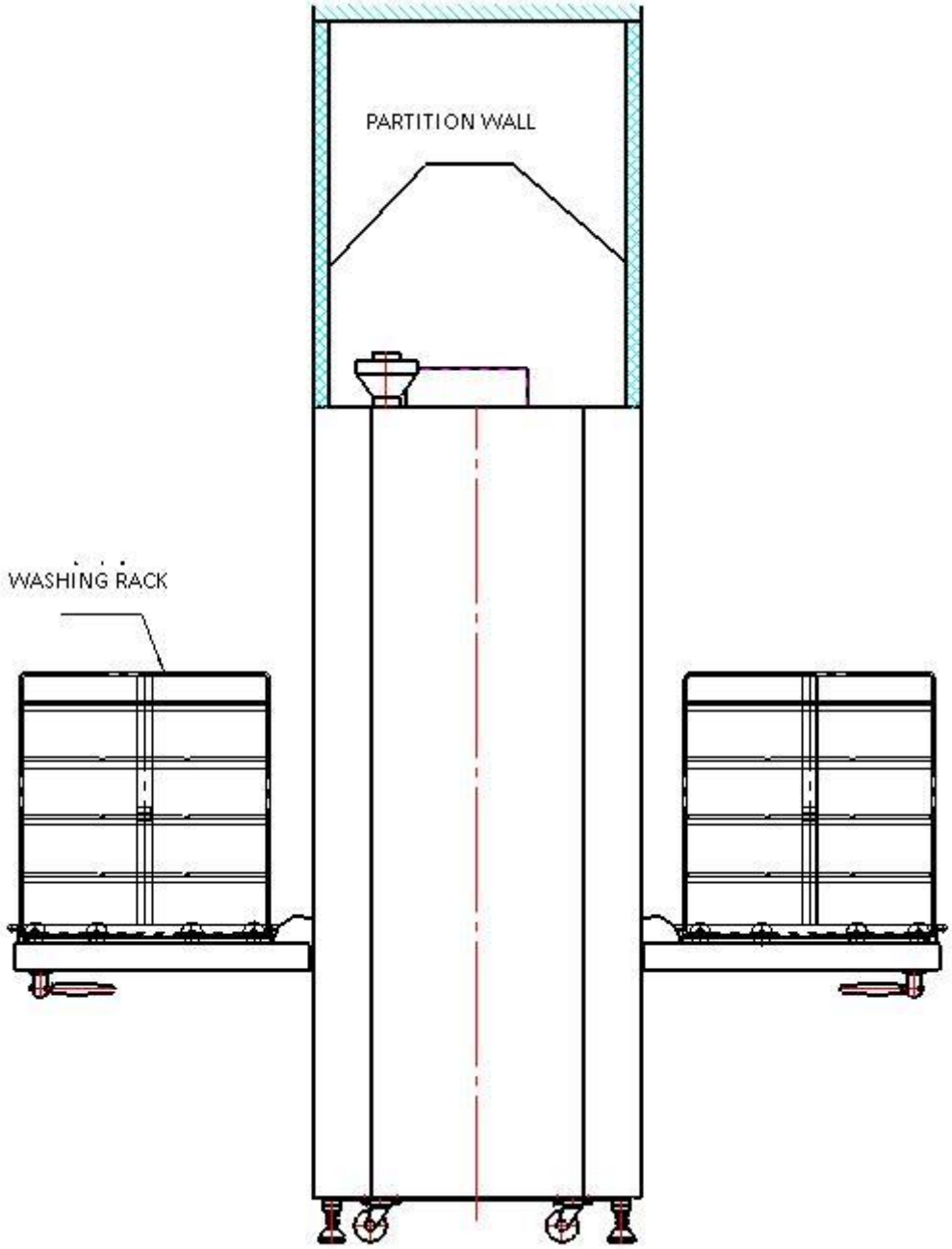


Fig 5

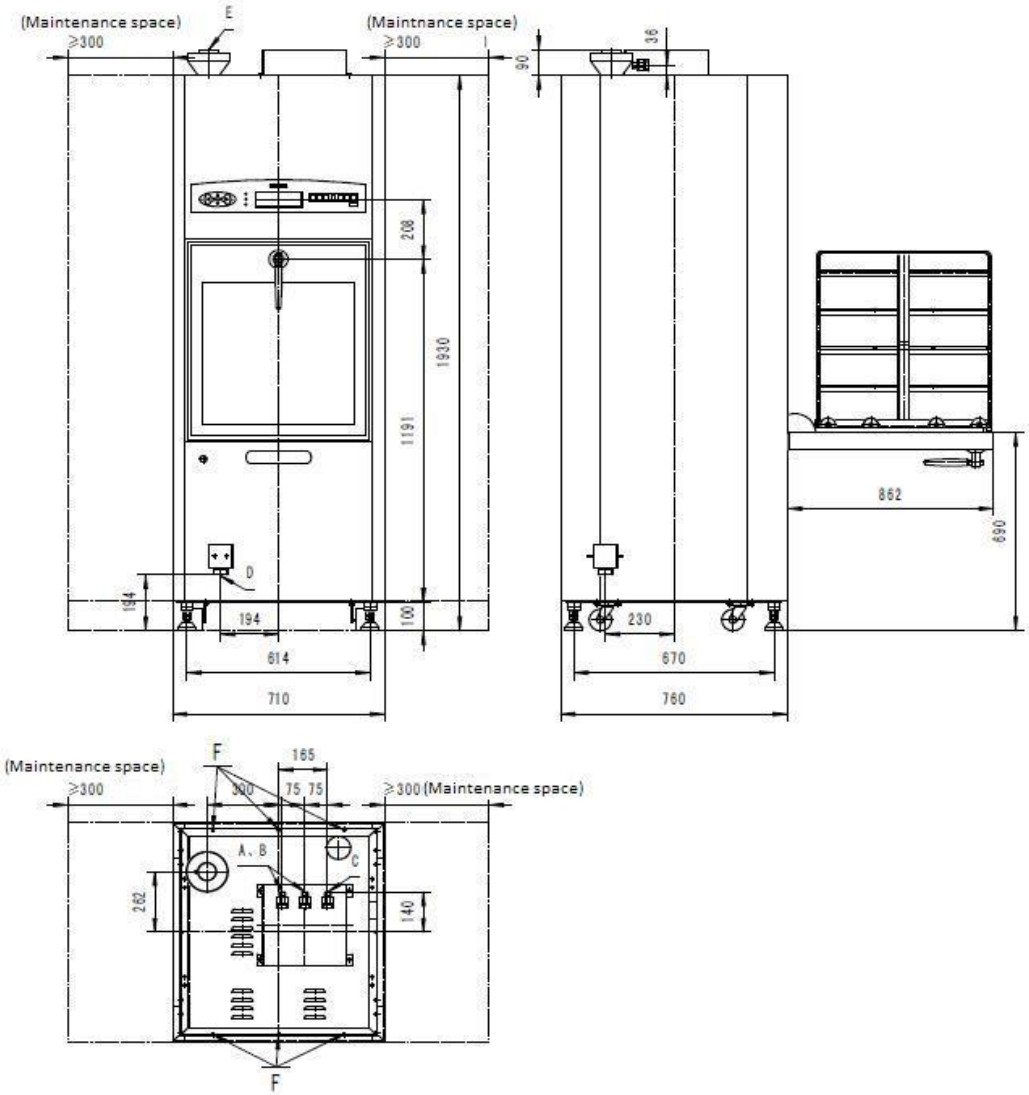
7. From the editor

1. This manual is as the guiding document, with the improvement of the products and technology progress, SHINVA reserves the right to amend it;
2. If user meets the situation that is difficult to operate according to this manual, such as the installation height is insufficient, the installation area is not enough and so on, please consult the technical engineers in charge of this project, and discuss a specific solution.
3. The specific technical parameters of the equipment can be referred to the technical parameter table.

Attachment 1：RAPID-M-320 Technical parameter table

Dimension	Washing chamber dimension/volume（L×W×H）mm/L		External dimension（L×W×H）mm		Washing rack dimension（L×W×H）mm		Equipment net weight（Kg）		Running weight（Kg）	
	601*592*987/320		760*710*1930		590*554*630		300Kg		360Kg	
Performance	Disinfecting temperature	Washing temperature	Disinfecting time		Running cycle (standard instrument program）		Dry temperature		Standard dry time	
	80℃～93℃	50℃	0～99 min continuously adjustable		41 min		70～110℃		20	
Max load capacity	Standard DIN instrument tray		Middle dressing bowl（16cm）		Middle kidney basin		Full volume			
	10 pieces（480×250×50）		36 pieces（reference）		48 pieces		320 Liter			
Energy requirement	Pure water		Drive power（electric heat）							
	Pressure MPa	Flowage	Voltage/Frequency	Power						
	0.2～0.5	1000Kg/hour	380/50HZ	22.5KW						
Energy cost (standard instrument program)	Saturated steam Kg/cycle		Pure water Kg/cycle		Cleaning liquid		Oiling liquid		Electricity（electric heat）	
	8		100		25000ml ×concentration ratio		25000ml ×concentration ratio		8.5 degree	
Joint dimension and drain capacity	**Water inlet**		**Drainage joint**	**Exhaust joint**	Waste water drainage capacity/temperature		Waste gas exhaust temperature			
	1/2 inch		**ϕ50**	Φ63	100Kg/20～100℃		50 ～ 100℃(normal pressure)			
Instruction	1.The equipment is electric heat type。 2. According to the different cleaning liquid /brand chosen by users from different manufacturers, its usage content is also different, so the detailed concentration ratio should read the reference from the manufacturer. 3. The equipment loading capacity is theoretical amount, only for reference, and it may be different according to the user location mode, space and other detailed situation. 4.According to the different equipment model, program selection procedures , and detailed user equipment for cleaning items, the energy consumption may be different									

Attachment 2：Installation drawing



Pipe joint table

Code	A, B	C	D	E
Name	Soft/pure water in	Pure water in	Drainage	Exhaust
Joint type	Outer screw	Outer screw	Outer screw	
Joint size	1/2"	1/2"	1 1/2"	φ90—130

Technical specification

1、Unit in drawing: mm

2、Installation requirement:

a、When multi equipments are installed in a line, between each one preserve space 300mm-500mm for matenance, and the maintenance door is in side of pollution area.

b、Partition wall can not be fixed on the equipment in any way.before partition.dismantle the srew in the postion F

c、Installtion must accord to the reqirement of installation manual

3. User supporting faciclities

3.1. Drive power:380V/3phase/22.5kw, control power: 220V/1phase, 0.5kw

3.2. Water source requirement: 02.2-0.5Mpa, flowage:≥25L/min, usage:30L/sage, pure water conductivity: ≤15μS/cm(25℃)

4. Ohters

4.1 Floor drain must be installed on the grond, at least one floor drain for each euipment

4.2 In the top of ther sterilizer, set a exhaust port (recomend), good for heat dissipation

4.3 Draing/exhaust temperature: ≤93℃

5. Equipment weight: 300kg/empty weight

Rapid-M-320	Fast-type fully-automatic washer-disinfector
Installation drawing	(20090820)
Shandong Xinhua Medical Instrument Co.,Ltd.	